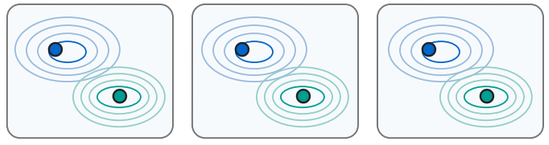
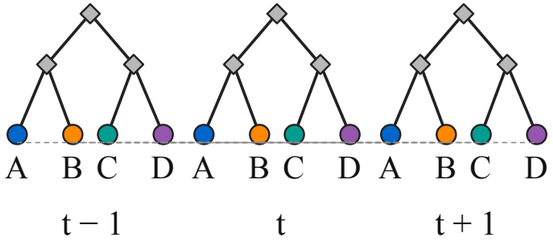
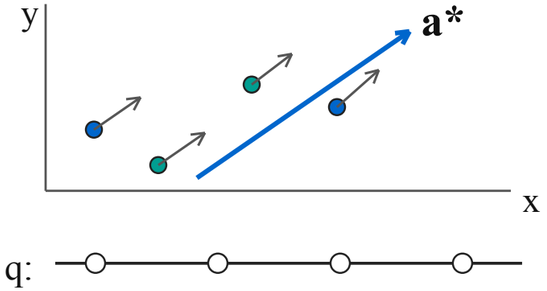
1 Extract & reference

Time-varying scalar fields
(2D / 3D, t − 1, t, t + 1)



Trees + track IDs



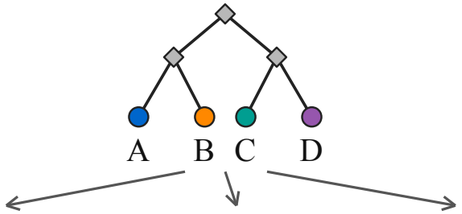


$q_i^t = \langle a^*, p_i^t \rangle - o$

One fixed spatial reference

2 Search leaf orders

Topology budget ε
$\eta = (\varepsilon_0 + \varepsilon)\,\Delta F$

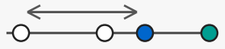




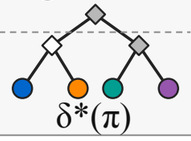
Width-aware refinement

Evaluate (fixed order π)

Displacement τ



Merge error δ*



δ*(π) ≤ η



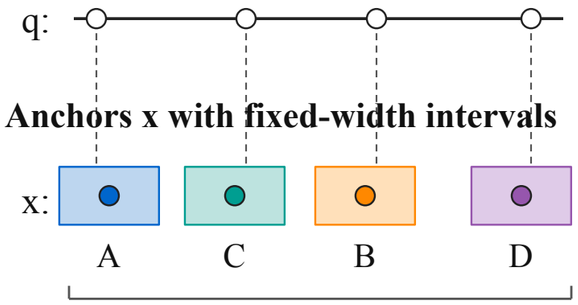
Select by (τ, NROE)



Finite heuristic search

3 Place anchors

Reference targets q (fixed reference)



Displacement cap: $\tau(\pi) + \beta$ o

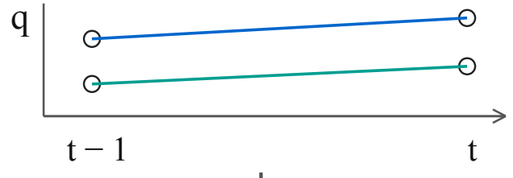
QP (selected order only)

$\min \Sigma_k \lambda_k E_k,\ k \in \{R, G, M, C\}$

Reference | Geometry | Motion | Centering

↑ Valid history

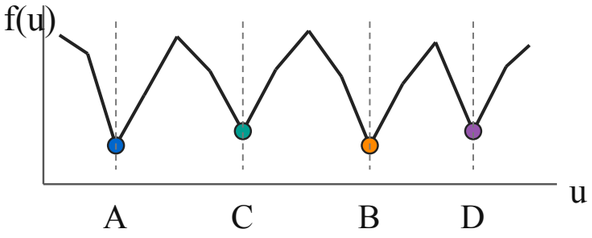
Matched projected motion



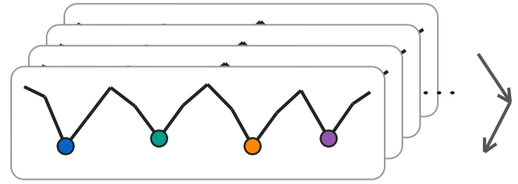
Anchors x + intervals

4 Reconstruct & assemble

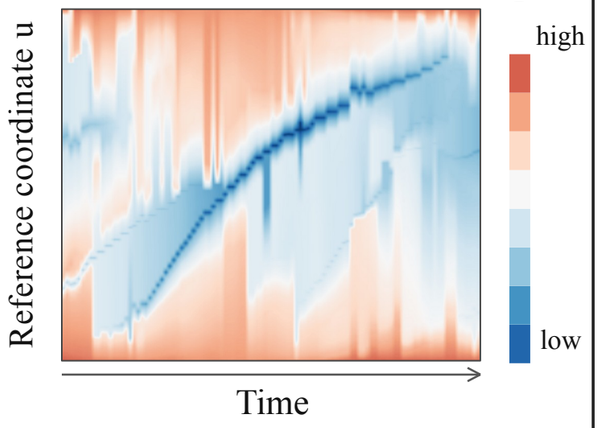
Reconstructed 1D scalar profile (join)



Leaf / LCA filling **Barrier correction**



Temporal merge tree map



Shared origin, direction & scale